

VIJAYA VITTALA INSTITUTE OF TECHNOLOGY



Department of Basic Science

QUESTION BANK

Subject Code: 1BMATE101

Subject: Differential Calculus & Linear Algebra

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For I Year B.E

Prepared By:
Mr.Ramesh

MODULE - 1

DIFFERENTIAL CALCULUS

- Q1.** With usual notations, prove that $\tan \phi = r \frac{d\theta}{dr}$.
- Q2.** Find the angle between the curves $r = 6 \cos \theta$ and $r = 2(1 + \cos \theta)$.
- Q3.** Find the angle between the curves $r = a(1 - \cos \theta)$ and $r = 2a \cos \theta$.
- Q4.** Find the angle between the curves $r = a(1 + \sin \theta)$ and $r = a(1 - \sin \theta)$.
- Q5.** Show that the curves $r = a(1 + \cos \theta)$ and $r = a(1 - \cos \theta)$ cut each other orthogonally.
- Q6.** Find the angle between the curves $r = a \log \theta$ and $r = \frac{a}{\log \theta}$.
- Q7.** Find the angle between the curves $r = \frac{a\theta}{1 + \theta}$ and $r = \frac{a}{1 + \theta^2}$.
- Q8.** Find the pedal equation of the curve $r^n = a^n \cos n\theta$.
- Q9.** Find the pedal equation of the curve $r^2 = a^2 \sec 2\theta$.
- Q10.** Find the pedal equation of the curve $\frac{\ell}{r} = 1 + e \cos \theta$.
- Q11.** Find the pedal equation of the curve $\frac{2a}{r} = 1 + \cos \theta$.
- Q12.** Find the pedal equation of the curve $r^m = a^m(\cos m\theta + \sin m\theta)$.
- Q13.** Find the radius of curvature for the curve $x^3y = a(x^3 + y^3)$ at $(-2a, 2a)$.
- Q14.** Find the radius of curvature for the curve $y = 1 + \sin x - \sin 2x$ at $x = \frac{\pi}{2}$.
- Q15.** Find the radius of curvature of the curve $\sqrt{x} + \sqrt{y} = 4$ at the point where it cuts the line through the origin making an angle 45° with the x -axis.
- Q16.** Show that for $r^n = a^n \cos \theta$, the radius of curvature varies inversely as r^{n-1} .
- Q17.** Show that for the catenary $y = a \log(\sec(x/a))$, the radius of curvature is $a \sec(x/a)$.
- Q18.** Find the radius of curvature for the curve $y^2 = a^2 \frac{(a-x)}{x}$ at the point where it meets the x -axis.
- Q19.** Show that for the curve $r(1 - \cos \theta) = 2a$, the square of the polar slope ϕ^2 varies as r^3 .
- Q20.** Show that for the curve $r = a(1 + \cos \theta)$, the quantity $\frac{p^2}{r}$ is constant.

MODULE - 2

POWER SERIES, INDETERMINATE FORMS AND MULTIVARIABLE CALCULUS

Q1. Expand $\log(\sec x)$ in powers of x up to the term containing x^4 using Maclaurin's series.

Q2. Expand $\log(1 + \cos x)$ as a Maclaurin series up to the term containing x^4 .

Q3. Using Maclaurin's series, expand $\sqrt{1 + \sin 2x}$ up to the term containing x^4 .

Q4. Expand $\log(\sec x)$ up to the term containing x^6 using Maclaurin's series.

Q5. Evaluate $\lim_{x \rightarrow 0} \left(\frac{a^x + b^x}{2} \right)^{\frac{1}{x}}$.

Q6. Evaluate $\lim_{x \rightarrow 0} (\tan x)^{\tan x}$.

Q7. Evaluate $\lim_{x \rightarrow a} \left(2 - \frac{x}{a} \right)^{\tan\left(\frac{\pi x}{2a}\right)}$.

Q8. Evaluate $\lim_{x \rightarrow 0} \left(\frac{a^x + b^x + c^x + d^x}{4} \right)^{\frac{1}{x}}$.

Q9. If $u = f(x - y, y - z, z - x)$, show that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$.

Q10. If $u = f\left(\frac{y-x}{xy}, \frac{z-x}{xz}\right)$, prove that $x^2 \frac{\partial u}{\partial x} + y^2 \frac{\partial u}{\partial y} + z^2 \frac{\partial u}{\partial z} = 0$.

Q11. If $u = f\left(\frac{x}{y}, \frac{y}{z}, \frac{z}{x}\right)$, prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 0$.

Q12. If $u = e^{ax+by} f(ax - by)$, prove that $b \frac{\partial u}{\partial x} + a \frac{\partial u}{\partial y} = 2abu$.

Q13. If $x + y + z = u$, $y + z = v$, $z = uvw$, find $\frac{\partial(x, y, z)}{\partial(u, v, w)}$.

Q14. If $u = x + 3y^2 - z^3$, $v = 4x^2yz$, $w = 2z^2 - xy$, find $\frac{\partial(u, v, w)}{\partial(x, y, z)}$ at $(1, -1, 0)$.

Q15. If $x = r \sin \theta \cos \phi$, $y = r \sin \theta \sin \phi$, $z = r \cos \theta$, find $\frac{\partial(x, y, z)}{\partial(r, \theta, \phi)}$.

Q16. Examine the function $f(x, y) = x^3 y^2 (1 - x - y)$ for extreme values.

Q17. Find the maximum and minimum values of $f(x, y) = x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$.

Q18. Examine the function $f(x, y) = xy(a - x - y)$ for extreme values.

MODULE - 3

ORDINARY DIFFERENTIAL EQUATIONS

Q1. Solve $(x^2 + y^2 + x) dx + xy dy = 0$.

Q2. Solve $(xy^3 + y) dx + 2(x^2y^2 + x + y^4) dy = 0$.

Q3. Solve $(x^2y^2 - e^{1/x^3}) dx - x^2y dy = 0$.

Q4. Solve $(xy + y^2 + y) dx + (x^2 + 3xy + 2x) dy = 0$.

Q5. Solve $(x^2 + y^2) dx + 2xy dy = 0$ using an integrating factor.

Q6. Solve $(x + y) dx + (x - y) dy = 0$ by finding a suitable integrating factor.

Q7. Solve $\frac{dy}{dx} + \frac{y}{x} = y^2x$.

Q8. Solve $\frac{dy}{dx} + y \tan x = y^3 \sec x$.

Q9. Solve $x \frac{dy}{dx} + y = x^3y^6$.

Q10. Solve $\frac{dy}{dx} + \frac{y \cos x + \sin y + y}{\sin x + x \cos y + x} = 0$.

Q11. Find the orthogonal trajectories of $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, where a is a parameter.

Q12. Find the orthogonal trajectories of $\frac{x^2}{a^2} + \frac{y^2}{b^2 + \lambda} = 1$, where λ is the parameter.

Q13. Find the orthogonal trajectories of $y^2 = 4ax$, where a is the parameter.

Q14. Find the orthogonal trajectories of the cardioid $r = a(1 - \cos \theta)$.

Q15. Show that the family of parabolas $y^2 = 4a(x + a)$ is self-orthogonal.

Q16. Solve $xp^2 - (x^2 + y^2)p + xy = 0$.

Q17. Solve $xyp^2 - (x^2 + y^2)p + xy = 0$.

Q18. Solve for p : $\frac{dy}{dx} \frac{dx}{dy} = \frac{x}{y} - \frac{y}{x}$.

Q19. Find the general and singular solutions of $xp^2 + xp - yp + 1 - y = 0$.

Q20. Solve the Clairaut's equation $(y - px)(p - 1) = p$.

Q21. Solve $(px - y)(py + x) = 2p$ by reducing it to Clairaut's form using $X = x^2$, $Y = y^2$.

Q22. Solve $(px - y)(py + x) = a^2p$ by reducing it to Clairaut's form using $X = x^2$, $Y = y^2$.

Q23. Show that the current i in an L - R circuit satisfies

$$L \frac{di}{dt} + Ri = E \sin \omega t$$

and find $i(t)$ given $i(0) = 0$.

Q24. An inductance of 2 henries and resistance of 20 ohms are connected in series with an emf of 100 volts; find the current at the end of 0.01 seconds.

MODULE - 4

ORDINARY DIFFERENTIAL EQUATIONS OF HIGHER ORDER

Q1. Solve $\frac{d^3y}{dx^3} + y = 0$.

Q2. Solve $(4D^4 - 4D^3 - 23D^2 + 12D + 36)y = 0$.

Q3. Solve $(D^4 + 18D^2 + 81)y = 0$.

Q4. Solve $(D^3 + 8)y = 0$.

Q5. Solve $(4D^4 - 8D^3 - 7D^2 + 11D + 6)y = 0$.

Q6. Solve $y'' + 3y' + 2y = 12x^2$.

Q7. Solve $y'' + 3y' + 2y = \cos^2 x$.

Q8. Solve $y'' - 4y = 3^x + \cosh(2x - 1)$.

Q9. Solve $(D^2 + 2D + 1)y = \sin 2x$.

Q10. Solve $y'' + 4y = x^2 + 2x$.

Q11. Solve $\frac{d^3y}{dx^3} + 2\frac{d^2y}{dx^2} + \frac{dy}{dx} = e^{-x} + e^x$.

Q12. Solve by variation of parameters $y'' - 2y' + y = e^x \log x$.

Q13. Solve by variation of parameters $y'' + a^2y = \sec ax$.

Q14. Solve by variation of parameters $y'' + y = \frac{1}{1 + \sin x}$.

Q15. Solve by variation of parameters $y'' + y = \sec x \tan x$.

Q16. Solve by variation of parameters $y'' + y' - 2y = \frac{1}{1 - e^{-x}}$.

Q17. Solve $x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + y = 2 \cos^2(\log x)$.

Q18. Solve $x^2 \frac{d^2 y}{dx^2} - 2x \frac{dy}{dx} - 4y = x^4$.

Q19. Solve $x^2 \frac{d^2 y}{dx^2} + 5x \frac{dy}{dx} + 4y = x^4$.

Q20. Solve $x^3 \frac{d^3 y}{dx^3} + 3x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + 8y = 65 \cos(\log x)$.

Q21. Solve $(3 + 2x)^2 \frac{d^2 y}{dx^2} + 5(3 + 2x) \frac{dy}{dx} + y = 4x$.

Q22. Solve $(1 + x)^2 \frac{d^2 y}{dx^2} + (1 + x) \frac{dy}{dx} + y = 4 \cos[\log(1 + x)]$.

Q23. Solve $(1 + x)^2 \frac{d^2 y}{dx^2} + (1 + x) \frac{dy}{dx} + y = \sin(2 \log(1 + x))$.

Q24. Solve $\frac{d^2 y}{dx^2} - 4 \frac{dy}{dx} + 5y = 0$, given $\frac{dy}{dx} = 2$, $y = 1$ at $x = 0$.

Q25. Solve the differential equation of an L-C-R circuit $L \frac{d^2 q}{dt^2} + R \frac{dq}{dt} + \frac{q}{C} = E$, where L , R , and C are constants.

MODULE - 5

LINEAR ALGEBRA

Q1. Find the rank of

$$A = \begin{bmatrix} 2 & -1 & -3 & -1 \\ 1 & 2 & 3 & -1 \\ 1 & 0 & 1 & -1 \\ 0 & 1 & 1 & -1 \end{bmatrix}.$$

Q2. Find the rank of

$$\begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}.$$

Q3. Find the rank of

$$\begin{bmatrix} 11 & 12 & 13 & 14 \\ 12 & 13 & 14 & 15 \\ 13 & 14 & 15 & 16 \\ 14 & 15 & 16 & 17 \end{bmatrix}.$$

Q4. Find the rank of

$$\begin{bmatrix} 1 & 3 & -1 & 2 \\ 0 & 11 & -5 & 3 \\ 2 & -5 & 3 & 1 \\ 4 & 1 & 1 & 5 \end{bmatrix}.$$

Q5. Test the consistency and solve:

$$\begin{cases} x - 4y + 7z = 14, \\ 3x + 8y - 2z = 13, \\ 7x - 8y + 26z = 5. \end{cases}$$

Q6. For what values of a and b does

$$\begin{cases} x + y + z = 6, \\ x + 2y + 3z = 12, \\ x + 2y + az = b \end{cases}$$

have (i) no solution, (ii) unique solution, (iii) infinite solutions?

Q7. Investigate λ, μ for which

$$\begin{cases} x + y + z = 6, \\ x + 2y + 3z = 10, \\ x + 2y + \lambda z = \mu \end{cases}$$

has (i) unique solution, (ii) infinite solutions, (iii) no solution.

Q8. Apply Gauss elimination to solve:

$$\begin{cases} 2x + 5y + 7z = 52, \\ 2x + y - z = 0, \\ x + y + z = 9. \end{cases}$$

Q9. Solve using Gauss elimination:

$$\begin{cases} 2x + y + 4z = 12, \\ 4x + 11y - z = 33, \\ 8x - 3y + 2z = 20. \end{cases}$$

Q10. Solve by Gauss elimination:

$$\begin{cases} x + 2y + 3z = 14, \\ 2x - y + z = 3, \\ 3x + y - 2z = 7. \end{cases}$$

Q11. Solve by Gauss–Seidel Method (3 iterations):

$$\begin{cases} x + y + 54z = 110, \\ 27x + 6y - z = 85, \\ 6x + 15y + 2z = 72. \end{cases}$$

Q12. Using Gauss–Seidel, solve:

$$\begin{cases} 27x + 6y - z = 85, \\ 6x + 15y + 2z = 72, \\ x + y + 54z = 110. \end{cases}$$

Carry out three iterations with initial guess $(0, 0, 0)$.

Q13. Solve by Gauss–Seidel Method:

$$\begin{cases} 5x + 2y + z = 12, \\ x + 4y + 2z = 15, \\ x + 2y + 5z = 20. \end{cases}$$

Use initial guess $(1, 0, 3)$ and 3 iterations.

Q14. Solve by Gauss–Seidel Method:

$$\begin{cases} 7x + 52y + 13z = 104, \\ 3x + 8y + 29z = 71, \\ 83x + 11y - 4z = 95. \end{cases}$$

Carry out 4 iterations.

Q15. Find eigenvalues and eigenvectors of

$$A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}.$$

Q16. Find eigenvalues and eigenvectors of

$$A = \begin{bmatrix} 1 & 2 & 2 \\ 0 & 2 & 1 \\ -1 & 2 & 2 \end{bmatrix}.$$

Q17. Find eigenvalues and eigenvectors of

$$A = \begin{bmatrix} 4 & 0 & 1 \\ 0 & 1 & 0 \\ -3 & -2 & 0 \end{bmatrix}.$$

Q18. Find eigenvalues and eigenvectors of

$$A = \begin{bmatrix} 3 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Q19. Find the largest eigenvalue and eigenvector of

$$A = \begin{bmatrix} 4 & 1 & -1 \\ 2 & 3 & -1 \\ -2 & 1 & 5 \end{bmatrix}$$

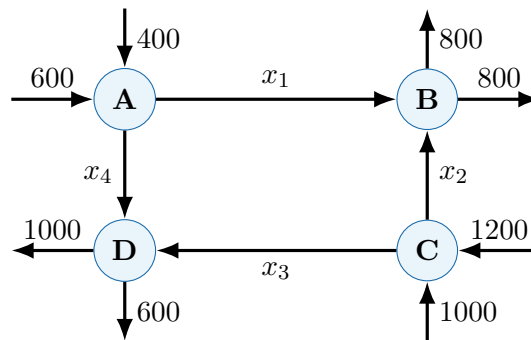
using initial vector $[1, 0.8, -0.8]^T$. Perform 5 iterations.

Q20. Using Rayleigh's Power Method, find dominant eigenvalue and eigenvector of

$$A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$$

taking $[1, 1, 1]^T$ as initial vector. Perform 6 iterations.

Q21. Write the system of linear equations of total flow in the network of one-way streets shown in the figure and find its solution.



Q22. Find the traffic flow pattern for the network shown in the following figure.

